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Palaeomagnetism of Silurian sediments in W Ireland: evidence for block rotation in the Caledonides

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SUMMARY

A palaeomagnetic study of mid-Silurian (Wenlock) sediments of the Salrock Formation in NW Galway yields a reliable Silurian palaeomagnetic pole position for W. Ireland at latitude 1.6°S, longitude 287.7°E (dp = 2.5°, dm = 4.6°, $N = 27$ sites).

A further 12 sampling sites were located in Silurian redbeds on Clare Island, S. Mayo where a palaeomagnetic remanence of Silurian age was also identified. Both remanences pre-date local late-Caledonian folding. However the Clare Island result has an anomalous azimuth when compared with the Salrock result and with published Silurian results for Britain. The two Irish results can be reconciled by anticlockwise block rotation of the Clare Island region through 40°–50°. The rotation brings both palaeomagnetic and structural trends within the two sampling regions into alignment. Strike-slip displacement across the Clew Bay fault zone, producing a sinistral shear couple, might have enabled such a rotation. The timing and geographic extent of the proposed block rotation is as yet poorly defined. However a younger age limit for rotation is provided by a late Devonian or Carboniferous partial magnetic overprint observed throughout both study areas, and from geological evidence it seems likely that rotation is confined to the Clew Bay fault zone.

The Dalradian rocks of the Connemara Massif, located to the south of the Salrock district, have an E–W structural grain. Some workers have proposed clockwise rotation of the Connemara Massif of *ca.* 30° at between 500 and 390 Ma which would bring the structural trends in W. Ireland into parallelism with Caledonian trends in Scotland. Because the Silurian Salrock succession (which rests unconformably on the Dalradian basement of Connemara) shows no such anomaly, it provides a younger age limit for the possible rotation of Connemara.

Key words: palaeomagnetism, Caledonian orogeny, block rotation, strike-slip tectonics

1 INTRODUCTION

Palaeomagnetism has recently proved to be a powerful method of analysing the tectonic evolution of orogenic belts (see Van der Voo & Channell 1980). More complete interpretation of complicated palaeomagnetic/structural problems has been possible through greater integration of the two disciplines (for example see Courtillot *et al.* 1986). Palaeomagnetism, therefore, is playing a more prominent role in the understanding of (for example) deformed continental margins.

The nature of relative plate motions will inevitably control the style of tectonism at the plate margins. This raises the possibility that anomalous palaeomagnetic results might give indication of plate convergence directions. The results presented in this paper define localised block rotation which is consistent with the recently proposed three continent model (Cocks & Fortey, 1982; Soper & Hutton, 1984), involving oblique sinistral convergence between the Laurentian plate and a plate with, as yet, poorly defined margins located to the south (Ziegler 1982 map 1; Coney, Jones & Monger 1980).

Similar block rotations have been identified from palaeomagnetic evidence in the Mesozoic–Cenozoic western margin of the North American plate (see Beck *et al.* 1986, and references therein). Lambert & McKerrow (1978) concluded that the geological setting of the British Caledonides and North American Cordillera was similar also with respect to major fault type and pattern.

One of the aims of the present study is to improve the precision of the Siluro–Devonian palaeogeographic reconstruction for Britain by incorporating new palaeomagnetic data from W Ireland, presented in this paper, from the margin of the former Laurentian plate. The results help in the evaluation of the coherence of the W Irish Caledonian terrain since Siluro–Devonian times, and of its relation to other parts of Britain.

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The new palaeomagnetic data presented here are from Silurian sediments at two locations in W Ireland. The first study area is in the Salrock Formation in the vicinity of Salrock village, NW Galway, and the second is Clare Island, off the coast of S Mayo (Fig. 1). Both areas lie to the north of the Iapetus suture (Phillips, Shillinan & Murphy 1976; McKerrow 1982) and provide an opportunity to add to the currently limited number of reliable Siluro–Devonian pole positions for this part of the British Isles. For a recent summary of the most reliable Siluro–Devonian pole positions currently available for Britain see Livermore, Smith & Briden (1985).

Morris (1973) carried out some alternating field (AF) cleaning of samples from two sections through the Salrock Formation, a sequence consisting predominantly of red siltstones. Although there is reason to believe that his study failed to resolve the multicomponent NRM satisfactorily, it nevertheless revealed the potential of these rocks as ancient remanence carriers. Therefore the Salrock Formation has been resampled, and a further suite of samples taken from structurally more complicated strata of similar age on Clare Island. Complete progressive demagnetisation has been applied to isolate components of NRM effectively.

The Clare Island and Salrock regions lie 25 km apart, to the south and north respectively, of the Ordovician S Mayo Trough, an E–W trending tract of the non-metamorphic Caledonides. The trough is approximately 25 km wide (the distance between Salrock and Clare Island). It is bounded both to the north and (uniquely in the British Caledonides) to the south in Connemara by Dalradian rocks with E–W structural grain. The Silurian strata unconformably overlie both folded Ordovician sediments and Dalradian metamorphics, and obscure the structural relationship between the two. Leake *et al.* (1983) point out that there are three plausible models for the development of anomalously positioned Dalradian rocks in Connemara which are (1) that the Connemara schists were close to the South Mayo trough when sediment was accumulating; (2) Dalradian and Ordovician rocks have been juxtaposed by transcurrent movements (Dewey & Shackleton 1984; Hutton & Dewey

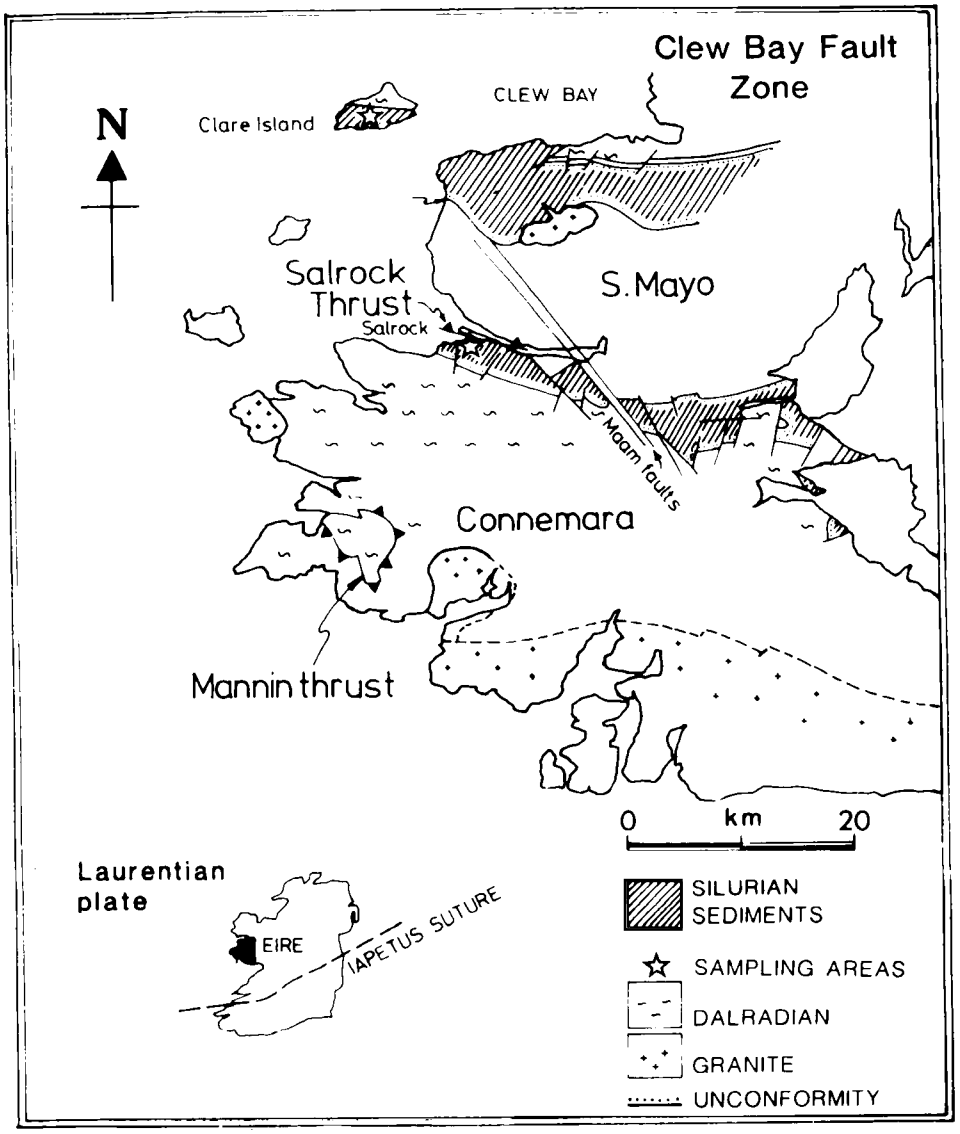


Figure 1. Simplified geological map of the Connaught area, W Ireland.

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1986; Bluck & Leake 1986); and (3) the Ordovician rocks have been thrust northwards over the Connemara region (also see Leake *et al.* 1983; Leake 1986).

There is clear evidence for sinistral displacements on lineaments in W Ireland from Odrovician to Devonian times, supporting the view that formerly remote terranes have become juxtaposed. Relative movement of terranes in post-Wenlock times could account for the absence of a local source of suitable detritus for the Clare Island succession at the present day (Phillips 1974; Phillips, Kennedy & Dunlop 1969; Phillips, Rickards & Dewey 1970). Relative movement between Clare Island and Salrock on tectonic breaks cannot be ruled out.

Leake *et al.* (1983) present convincing evidence for southward displacement and rotation of the Connemara Dalradian into its present east–west alignment by movement on the Mannin thrust. The age of displacement causing the clockwise rotation has been estimated using cross-cutting relationships at between *c.* 500 and 398 Ma, and by age determination on the deformed rocks in the thrust zone itself which give an age of 460 ± 25 Ma (Leake *et al.* 1983).

2 GEOLOGY OF THE SALROCK DISTRICT

The Silurian Salrock Formation lies 2.5 km to the north of the unconformity between Silurian strata and the Dalradian rocks of Connemara. Dating of these rocks is established through well preserved faunas as Wenlock (Phillips 1974; McKerrow & Campbell 1960; Carruthers & Muff 1909). The sampled succession (Fig. 2) consists of *c.* 828 m of interbedded red and green fine grained sandstones and siltstones.

The sediments host a number of andesite and lamprophyre sills, which are up to 5 m thick. There is clear field evidence for sill emplacement into incompletely lithified sediment, and McKerrow & Campbell (1960) report folding of similar intrusions nearby, confirming their pre-folding emplacement.

The strata generally dip at steep angles towards the north. There has been post-Silurian faulting in the Salrock district and further south in Connemara, with most major faults accommodating much larger displacements in the Ordovician than in the Silurian period (Ryan, pers. comm.). The Salrock thrust truncates the top of the Salrock Formation and is believed to have accommodated small displacements, with only minor rotational effects in its immediate vicinity.

2.1 The geological setting of sampling localities

Thirty-six sampling localities were drilled in a traverse through *c.* 750 m of the Salrock Formation, from within 6 m of the base near Salrock school to the uppermost

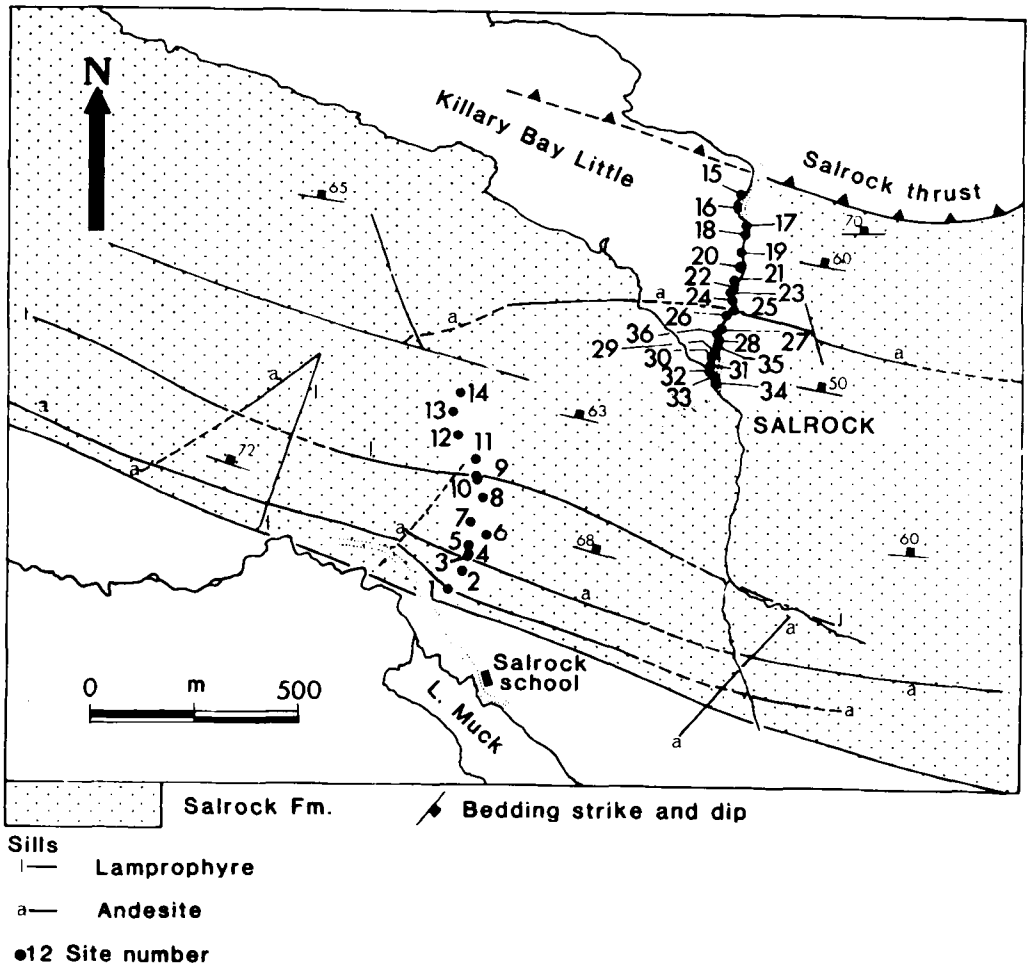


Figure 2. Geological map of the Salrock district showing the distribution of palaeomagnetic sampling sites.

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exposed strata 88 m from the Salrock thrust (Fig. 2). The sampling traverse is offset to follow good outcrop. Table 1 shows the general lithology and stratigraphic position of the rocks at all sampling localities in the Salrock Formation. The stratigraphically lower half of the transect covering the ground between sites S1 and S14 follows the course of Laird's sedimentary log (the first 365 m; Laird 1969), and the upper half (sites S34 to S15) adds a further 372 m to the total thickness sampled. The two halves of the section have average bedding strikes and dips of 279/64 N and 270/57 N respectively.

In two cases sampling of both sill (S3 and S9) and adjacent sediment provided potential contact tests of palaeomagnetic stability (sites S3 and S4, S9 and S10).

3 GEOLOGY OF CLARE ISLAND

The Silurian succession on Clare Island consists of at least 1700 m of red siltstones, sandstones and conglomerates resting unconformably on Dalradian and older basement. It records a cycle of marine transgression and regression within a marginal marine and fluviatile sedimentary environment.

The petrography of the clastics indicates a northerly sediment source which can no longer be identified to the north of Clare Island (Phillips 1974). Similarly, volcanic rocks on Clare Island provide evidence of northerly provenance but no source area survives. This can most readily be explained by strike-slip displacement on the Clew Bay fault zone and/or the Leck fault (Figs 1 and 3). Max & Riddihough (1975) and Max, Ryan & Indamdar (1983) suggest that the faults cutting across Ireland to emerge at Clew Bay are a continuation of the Highland Boundary Fault of Scotland. Local evidence from the head of Clew Bay for sinistral brittle–ductile shear has been described by Graham (1981) and sinistral strike-slip elsewhere in Ireland north of the Iapetus suture by Hutton (1982), Long & Max (1977), and also see Elders (1987).

In SW Mayo the strata can only be dated using poorly preserved faunas which yield an Upper Llandovery or Wenlock age. However the relative ages of the Salrock and Clare Island successions may be established with reasonable accuracy by matching major basin-wide facies changes. In this way Phillips (1974) has correlated the regressive Bunnamohaun Formation on Clare Island with the regressive cycle of sediments on the southern margin of the trough of which the Salrock Formation is the uppermost part. The rocks from both regions are, therefore, regarded as contemporaneous and probably Wenlock in age.

3.1 The geological setting of sampling localities

The approximate stratigraphic distribution of the 12 sampling localities on Clare Island is indicated in Table 3. The structural history of these rocks is more complicated than that of the Salrock Formation. Sampled strata may be divided into three structural domains: sites B1 to B7, and I1 and I2 structurally overlie the Owenmore thrust in openly folded redbeds and ignimbrite flows; both sites A1 and B8 lie in near vertically inclined strata below the Owenmore

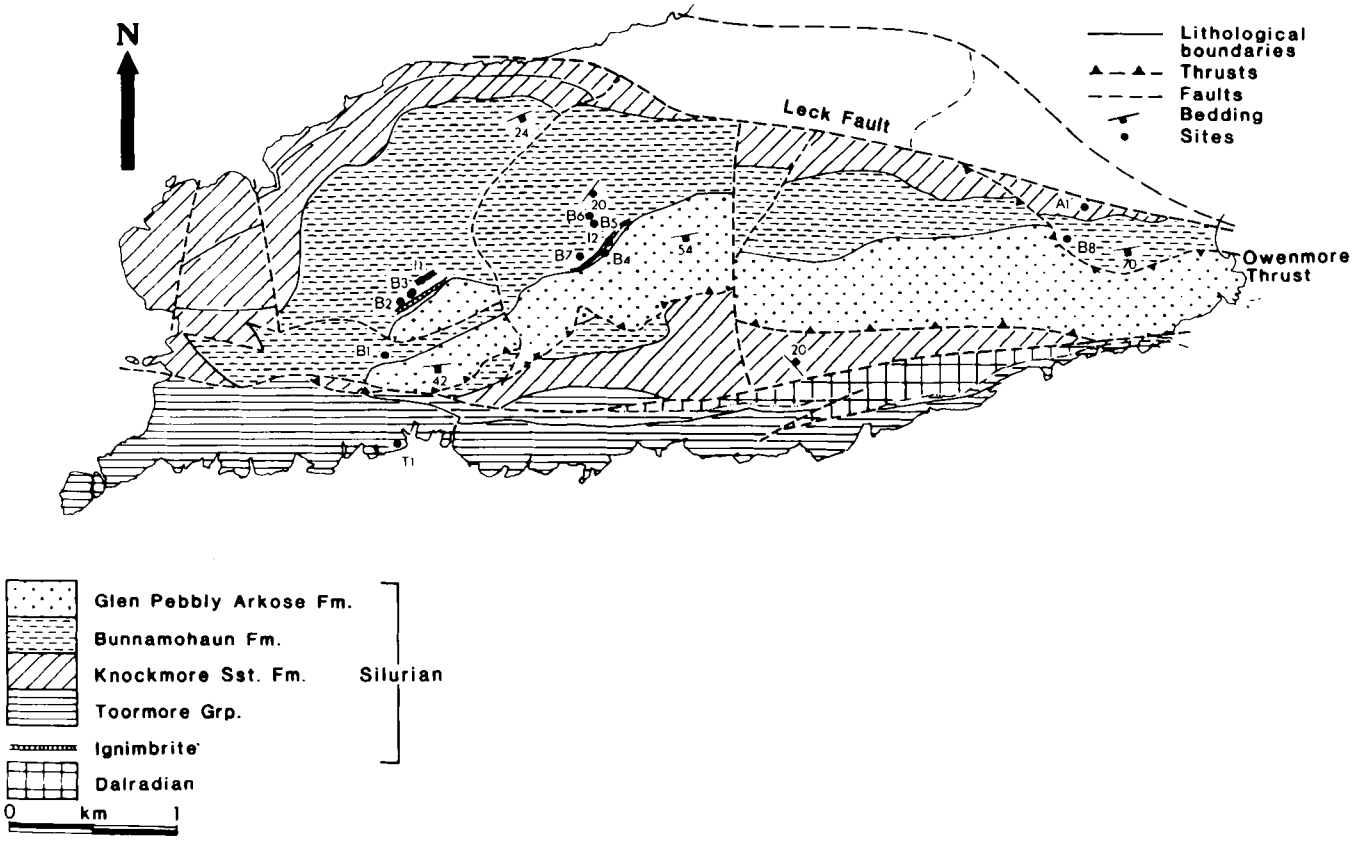


Figure 3. Simplified geological map of Clare Island showing palaeomagnetic sampling sites.

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thrust; A1 is near the Leck fault and B8 near the Owenmore thrust (Fig. 3); Site T1 lies both below the Owenmore thrust and to the south of the Kill fault, in the most deformed redbeds on Clare Island, which often exhibit penetrative cleavages.

4 PROGRESSIVE DEMAGNETISATION: SALROCK FORMATION SEDIMENTS

The distribution of NRM directions from the Salrock Formation given in Fig. 4 shows clear lithological control. Both sills and green sediments, with the exception of samples in the metamorphic aureole adjacent to the sills, are magnetised steeply downwards. Red siltstones and sandstones are generally magnetised in a northeasterly and shallow direction, and the red mudstones yield a crescent of NRM directions between that northeast shallow direction and southerly shallow directions. Sediments adjacent to sills yield westerly directed NRMs with varied downward inclinations. NRM intensities normally range between 1 and 10 mA m^{-1} in the redbeds, between 0.3 and 6 mA m^{-1} in green sediments, and between 5 and 200 mA m^{-1} in the intrusives.

A total of 198 samples has been progressively demagnetised in 12–24 steps, up to either 680 °C or 180 mT (peak). Of the demagnetised samples 125 were from red silt/sandstones, 51 from green silt/sandstones, and 22 from the andesite and lamprophyre sills.

Linear and planar elements in demagnetisation data were identified and their confidence limits defined using the LINEFIND statistical package (Kent, Briden & Mardia 1983).

AF and thermal demagnetisation effectively demagnetised green sediments and the igneous rocks, although AF treatment commonly failed to discriminate satisfactorily between vector components of NRM. On the other hand thermal treatment was found to be an effective means of isolating components of NRM, although heating often led to an increase of up to 600 per cent in bulk susceptibility of samples, particularly above 650 °C.

Problems with mineralogical alteration at high treatment temperatures are less pronounced in the red sediments. Any laboratory-generated magnetisations were identified during thermal demagnetisation by systematically reorienting samples during the series of treatments. Most redbed samples respond poorly to AF treatment, losing less than 20 per cent of their NRM by the maximum alternating fields available (180 mT), hence thermal demagnetisation has yielded the bulk of remanence directions from the red sediments.

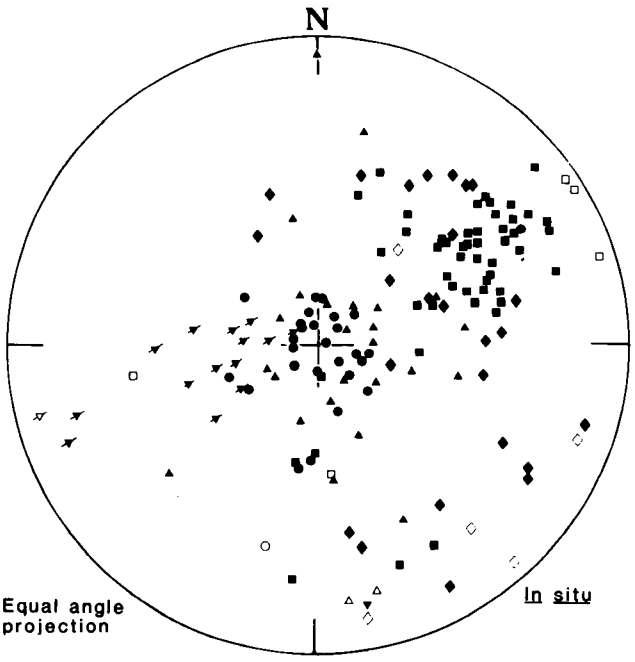


Figure 4. NRM directions of samples from the Salrock district in equal angle stereographic projection. Closed (resp. open) symbols lie in the lower (resp. upper) hemisphere. ■—red silt/sandstone, ◆—red silt/mudstone, ▲—green silt/sandstone, ●—andesite and lamprophyre, ⚡—contact metamorphics.

4.1 Components of NRM

Figure 5 consists of a selection of orthogonal plots typifying the behaviour of the major lithological groups upon demagnetisation. All vector components of NRM identified in the Salrock region can be assigned to one of three groups, labelled H, I, and L. Assignment to a group is generally made on the basis of blocking temperature and coercive force range (high, intermediate and low respectively). A small number of directions with atypical stability ranges from samples in unusual geological settings have been assigned to groups H, I and L through geological inference and similarities in remanence direction. It is not suggested that components which are given the same label necessarily all have the same mode of origin.

Site mean directions for components H and I (*in situ*) are shown in Fig. 6 and listed with radii of 95 per cent confidence in Table 1. Component L directions are steeply downwards towards present day north and are regarded as recent viscous remanent magnetisation (VRM). Component I directions lie at shallow inclinations towards the SSW, and component H directions are well grouped towards the ENE with shallow downwards inclinations (Fig. 6a). The distribution of NRM directions (Fig. 4), therefore, seems to stem from variations in the relative proportions of the above underlying characteristic remanent magnetisations preserved in each lithology.

4.2 Red sediments

The red sediments commonly preserve all three vector components of NRM (Fig. 5a–d). The thermal stability range of components L (up to 215 °C) and H (575 to 680 °C are narrow compared with that of component I (140–590 °C), yet component H usually constitutes the bulk of total NRM, with components L and I comprising approximately equal proportions. Component L is often removed after only two or three treatment steps and is consequently directionally ill-defined. This remanence is probably a VRM carried by multi-domain (MD) magnetite, which has been identified in thin section, and/or by mineral phases such as goethite.

The thermal stability range of component I extends beyond the Curie point of magnetite, and both magnetite

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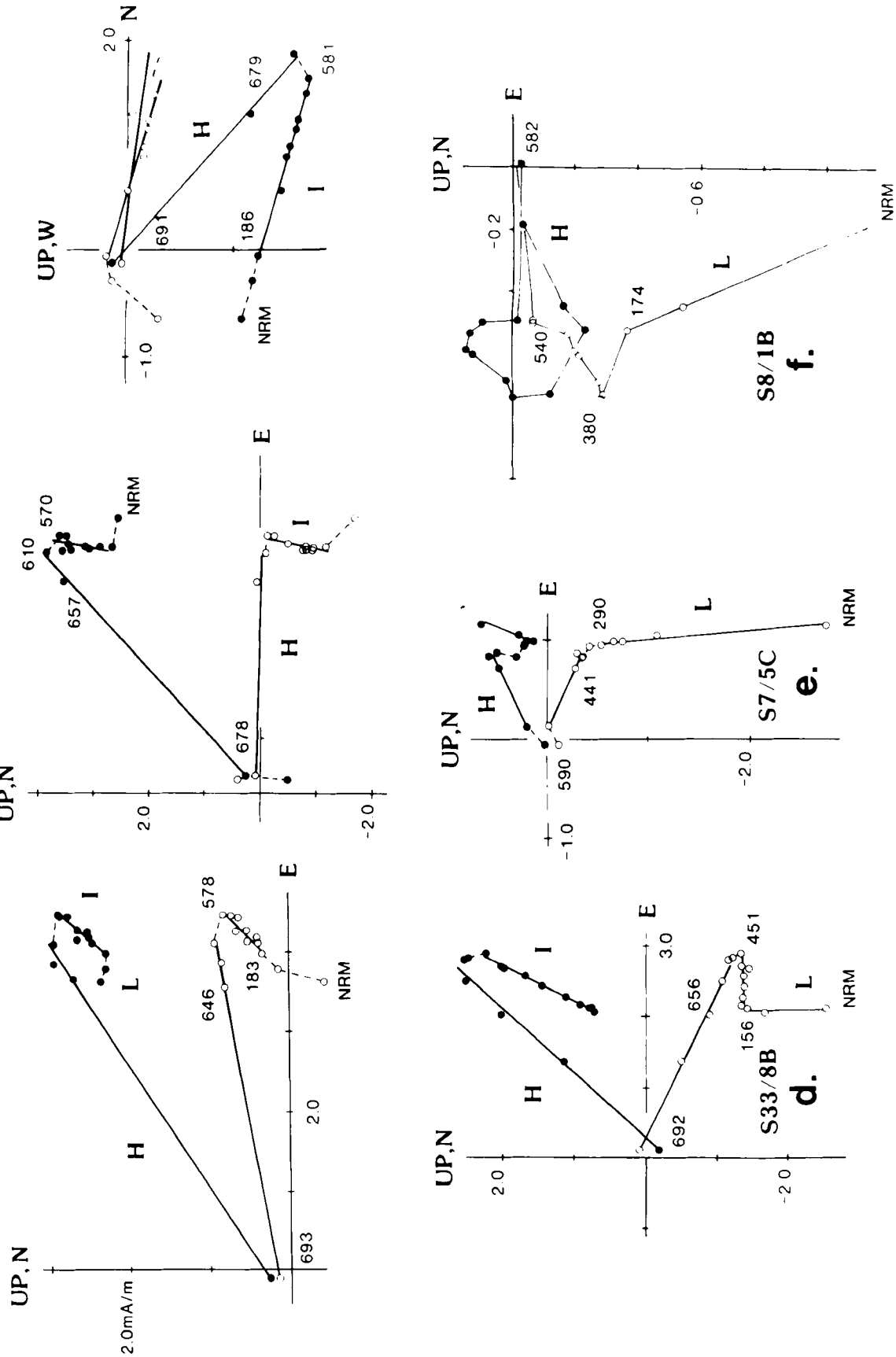


Figure 5. Orthogonal plots of typical progressive thermal demagnetisation results from sediments of the Salrock Formation. (a), (b) red siltstone; (c), (d) red silt/mudstone; (e), (f) green siltstone. Treatment temperatures are in °C and intensities of magnetisation in mA m^{-1}. Open (resp. closed) symbols lie in the vertical (resp. horizontal) projection, L, I and H refer to low, intermediate and high Tb NRM components respectively.

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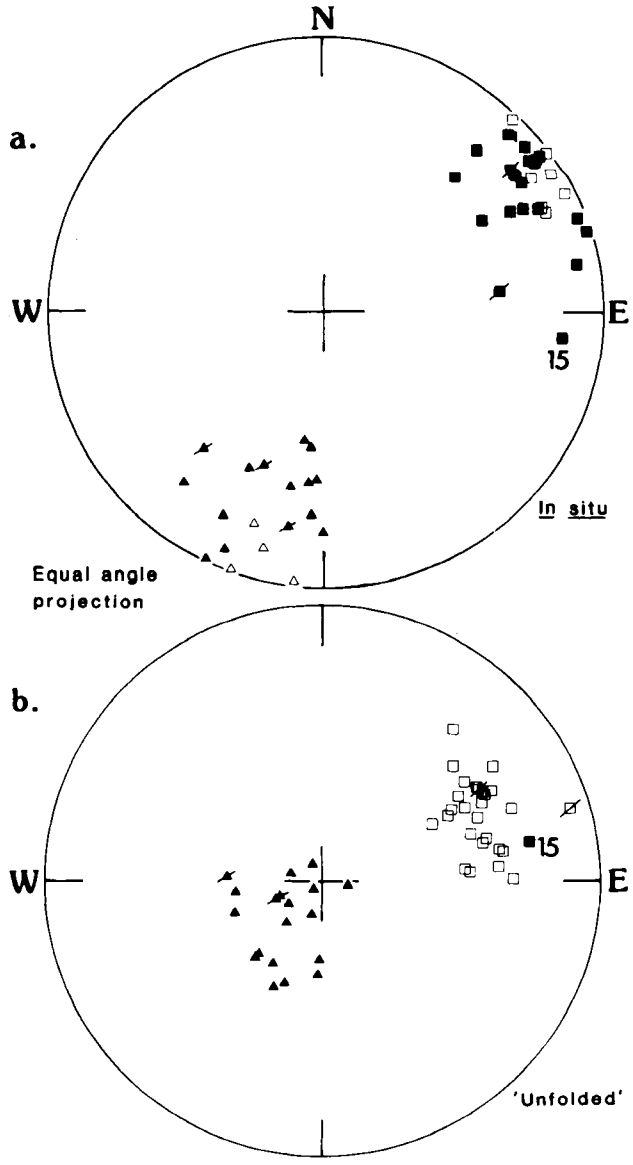


Figure 6. Site level mean directions of H(■) and I(▲) NRM components. (a) *In situ* directions. (b) 'Unfolded' directions. Symbols with an oblique stroke indicate results which are not statistically significant at the 95 per cent confidence level. Radii of circles of 95 per cent confidence about the statistically significant mean directions are listed in Tables 1 (for H) and 2 (for I). The H component result from site S15 is indicated.

and haematite carry this remanence. The mode of origin of component I is discussed in detail later.

Component H has blocking temperatures between the Curie points of magnetite and haematite, and is carried solely by haematite.

4.3 Green sediments

Green sediments rarely preserve all three components of NRM (Fig. 5e, f). The NRM persists up to 580 °C (approximately the Curie point of magnetite) demonstrating the effective absence of haematite in these samples. Component L is usually strong, with Tb up to 350 °C and is carried by an unidentified host mineral phase, possibly MD magnetite or goethite. Component I is less frequently observed and is removed between 200 and 540 °C, i.e. near to the Curie point of its host mineral phase, which is low titanium-magnetite. Magnetizations with typical component H directions are rarely isolated but when they are identified (Fig. 5e) they too are carried by magnetite, with blocking temperatures near to the Curie point. Reversed polarity component H directions have been observed at sites S7 and S8 (Fig. 5f).

5 PROGRESSIVE DEMAGNETIZATION: SILLS AND THEIR METAMORPHIC AUREOLES IN THE SALROCK FORMATION

5.1 The sills

The NRM of samples from andesite and lamprophyre intrusions (sites S1/3/25 and S9 respectively) consists almost entirely, with the exception of some samples from S25, of vector component L, scattered about the present Earth's field direction, with Tb up to 290–570 °C. A second poorly defined component of NRM has been identified at S3 and S25 which has a narrow Tb range from 500–560 °C to the Curie point of magnetite, constituting only a small fraction of the total NRM (Fig. 7a). This component has westerly declinations and shallow inclinations: a direction rarely observed in sedimentary lithologies away from the intrusions (Fig. 5f). There is only a suggestion of this component in a few samples from the other intrusions, also at high blocking temperatures. The direction of this component is ill-defined and sample directions do not yield a significant site level direction. The NRM is carried by multi- and single-domain magnetite. The characteristics of the NRM in the contact metamorphic rocks must be examined to shed some light on the origin of this westerly directed component.

5.2 The contact metamorphic rocks

It is evident from Fig. 7(b, c) that green sediments adjacent to the sill at site S4 have very different magnetic characteristics from those sampled elsewhere in the sequence (Fig. 5e, f).

One of the most striking features of the NRM of the green sediments adjacent to the sills (sites S4 and S10) is a well-defined component with a WSW declination and shallow negative inclination. This is stable between 360–450 °C and 650–680 °C and is carried by both magnetite and haematite. It is of reversed polarity (defined with respect to the apparent polar wander path for the British Isles). The WSW directed component has been identified at site S10 (Fig. 7e) but is more frequently observed at S4 where it is found to lie antiparallel to component H directions from samples away from the intrusion (Fig. 7d) or from other sites. Therefore, this remanence has been labelled −H, and is converted to normal polarity for the purposes of fold tests and statistical analyses in Table 2.

Not only is component −H reversed, but NRM intensities increase sharply towards the intrusion, and component −H directions are very precisely defined. Component I is not

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Table 1. Salrock Formation: site mean directions of components H and I (*in situ*)

SITE	LITHOLOGY	Strat. Posn.(m)	Strike, Dip(°)	Component H: D(°)	I(°)	N	k	α_{95}(°)	Component I: D(°)	I(°)	N	k	α_{95}(°)	SITE
THRUST-828														
S15	Red Sltst, Cleaved	747	263 73N	96.7	9.0	6	32.9	11.9	189.5	12.8	3	5.8	NS	S15
S16	Fractured Massive Red Sltst	719	269 63N	63.1	-2.0	5	57.4	10.2	220.7	23.8	3	6.1	NS	S16
S17	Red Sltst	696	268 53N	65.0	-8.0	5	66.6	9.4	182.1	23.3	4	16.0	23.7	S17
S18	''	685	264 56N	56.3	-6.9	5	33.9	13.3	189.2	38.2	2	66.7	31.1	S18
S19	Massive fine Red Sst	654	264 57N	44.3	22.9	5	9.0	27.0	190.5	24.5	6	7.7	25.8	S19
S20	Purple/Red Sltst/Sst	634	271 58N	63.5	9.0	7	12.5	17.8	203.1	14.5	1	-	-	S20
S21	Green Sst	618	266 60N	49.4	3.4	1	-	-	183.7	-27.0	1	-	-	S21
S22	Purple/Red Sltst	591	269 55N	53.4	3.7	5	218.4	5.2	201.3	27.0	3	6.5	NS	S22
S23	Massive Red Sltst	572	275 61N	63.7	-8.8	5	20.4	17.4	185.3	26.3	4	17.9	22.3	S23
S24	Purple/Red Sltst	553	270 61N	58.3	-2.3	5	35.1	13.1	185.0	36.7	4	108.9	8.8	S24
S25	Andesite (‖1.5m thick)	525	273 60N	-	-	-	-	-	-	-	-	-	-	S25
S26	Flaggy Red Sltst	505	265 55N	52.8	2.2	6	18.5	16.0	178.9	-0.5	2	8.3	NS	S26
S27	Massive Red Sltst	476	265 57N	45.5	4.2	4	70.1	11.1	214.4	25.9	5	9.7	25.9	S27
S36	Red and Grey/Green Sltst	467	271 55N	56.0	9.6	5	73.0	9.0	194.0	-7.0	4	11.7	28.0	S36
S28	Flaggy Red Sltst/Sst	458	266 41N	53.9	-0.6	5	71.0	9.2	180.0	12.4	2	92.8	26.2	S28
S35	'' ''	447	282 66N	61.3	16.1	5	66.0	9.5	230.4	11.4	1	-	-	S35
S29	'' ''	437	286 50N	51.9	3.3	4	269.0	5.6	198.3	-8.0	1	-	-	S29
S30	Red Sltst	416	275 57N	43.6	-1.4	5	227.8	5.1	218.9	13.1	5	8.5	27.8	S30
S31	Red Sltst with Mud Lamellae	409	270 59N	46.0	4.2	6	29.8	12.5	197.8	-12.3	6	167.8	5.2	S31
S32	'' ''	399	271 56N	50.4	4.5	5	103.1	7.6	198.5	-0.3	6	35.2	11.5	S32
S33	'' ''	389	268 56N	50.4	9.6	5	43.5	11.7	202.5	3.9	5	34.4	13.2	S33
S34	'' ''	375	263 57N	42.3	11.6	5	247.5	4.9	202.9	35.2	1	-	-	S34
S14	Pale Green Sst	366	269 72N	75.8	36.9	1	-	-	-	-	-	-	-	S14
S13	Red Sltst/Sst	338	270 68N	59.5	23.9	5	173.5	5.8	211.3	52.3	1	-	-	S13
S12	Red Sltst	304	269 57N	52.2	9.4	4	3.5	NS	205.5	0.2	6	6.3	28.9	S12
S11	Green Sst	256	278 70N	-	-	-	-	-	-	-	-	-	-	S11
S9	Lamprophyre sill (3.2m thick)	232	270 62N	-	-	-	-	-	-	-	-	-	-	S9
S10	Green Sltst (Met)	227	287 59N	83.8	26.1*	3	3.1	NS	205.8	10.9	7	8.5	22.0	S10
S8	Green Sst	173	282 59N	72.4	1.3	3	14.9	33.2	186.0	-0.6	5	12.0	23.0	S8
S7	Green/Grey Sst	141	265 63N	53.2	9.8	3	75.2	14.3	207.2	-45.3	1	-	-	S7
S6	Green Sltst	114	286 63N	-	-	-	-	-	190.5	12.6	2	8.1	NS	S6
S5	Mainly Red sltst	67	284 66N	61.4	-12.4	5	65.8	9.5	182.6	23.2	1	-	-	S5
S4	Red and Green Sltst/Sst (Met)	47	288 64N	68.8	1.8*	12	59.1	5.7	-	-	-	-	-	S4
S3	Andesite sill (1.6m thick)	45	281 60N	-	-	-	-	-	-	-	-	-	-	S3
S2	Red Sltst	33	283 72N	79.0	6.0	5	88.3	8.2	183.2	17.1	4	10.8	29.4	S2
S1	Andesite sill (0.9m thick)	6	292 59N	-	-	-	-	-	-	-	-	-	-	S1
	First redbed-0													

Sst= Sandstone, Sltst= Siltstone, Met= Metamorphosed, N= Number of observations, k= Fisher's precision estimate, α_{95}= Radius of the circle of 95% confidence, NS= Not significant (p=0.05), *= Mixed polarity.

present near the intrusion but component L is present in most samples, with Tb up to 400 °C.

Like the NRM, saturation remanence increases sharply towards the intrusion as does, to a lesser extent, coercivity of remanence. This is in accordance with increased alteration and assimilation of the country rocks toward the sill margin, which has resulted in overprinting of original magnetic characteristics. The most conspicuous effect of sill intrusion on the character of the country rocks is seen in the field in the form of local sediment dewatering, establishing a similarity between the ages of sedimentation and intrusion. The sediment has also been enriched in fine-grained magnetite near the sills. The blocking temperature divide between the two main NRM components −H and L increases only slightly towards the sill margin, while the magnitude of −H increases significantly. Component −H is probably, therefore, a CRM (chemical remanent magnetisation) imposed during sill intrusion.

The sill at S3 is a poor carrier of ancient remanence (coercivity of remanence = 0.025 T) compared with the metamorphosed rocks (coercivity of remanence = 0.09 T). The poorly defined westerly directed high Tb component identified in the sills themselves (Fig. 7a) is regarded as the vestige of primary TRM (thermoremanent magnetisation) of the same age and polarity as the better preserved −H CRM found in the metamorphosed country rocks. It is interesting to note that the Earth's field direction at the time of sill emplacement, apart from being of reversed polarity, is indistinguishable from the component H directions identified elsewhere in the Salrock Formation (Table 1).

6 FOLD TESTS IN THE SALROCK REGION

The sampled section in the Salrock Formation includes two internally uniform structural units: the strata in both units dip moderately steeply towards the north, with *ca.* 12° difference in average orientation. The stratigraphically lower unit occupies the ground between sites S1 and S14, and the upper unit between sites S34 and S16 (Fig. 2, Table 1). The strike and dip of bedding measured at each site are given in Table 1; and the 'folded' and 'unfolded' site mean component H and I directions are shown in Fig. 6(a, b). Component H directions from site S15 are excluded from the fold test analysis due to their apparent reorientation associated with local thrusting and cleavage development in addition to folding.

Fold axes in the Salrock region are assumed to be parallel to the strike of strata and a single rotation is used to restore beds to their original horizontal attitude. Both the widely used McElhinny (1964) and McFadden & Jones (1981) fold tests were applied to the component H and I site mean directions.

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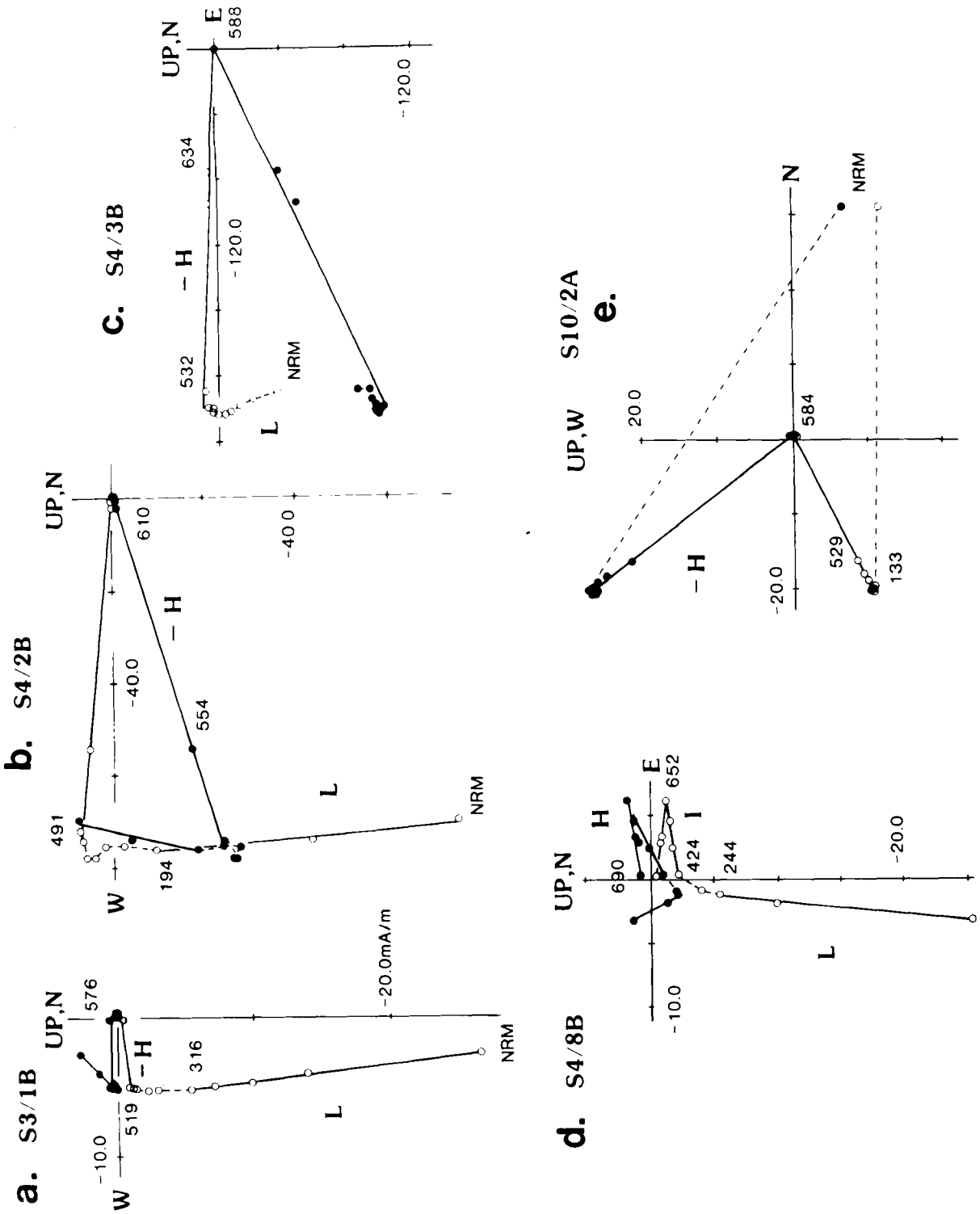


Figure 7. Orthogonal plots of typical results from progressive thermal demagnetisation experiments on samples from sills and nearby country rocks in the Salrock Formation. (a) andesite at site S3; (b), (c) green sandstone from site S4, 11 and 23 cm from the andesite sill at S3; (d) red siltstone 135 cm from the andesite sill, (e) green siltstone 10 cm from the lamprophyre sill at S9. Symbols and labels are as in Fig. 5 except −H indicates reversed polarity H remanence components.

Table 2. Salrock Formation: group of sites level statistics and fold tests

Component H

Limb1: Sites S2,4,5,7,8,12,13
Limb2: Sites S34 to S29,S35,S28,S36,S27,S26,S24 to S16

Limb1 statistics

Mean direction:	D(°)	I(°)	N	R	k	α_{95}(°)
in situ	63.9	9.4	7	6.857	42.00	9.4
"unfolded"	72.0	-25.6	7	6.811	31.75	10.9

Limb2 statistics

Mean direction:						
in situ	53.5	3.7	20	19.670	57.51	4.3
"unfolded"	64.3	-27.9	20	19.633	51.83	4.6

Overall statistics

Mean direction:							<-----Pole-----|| Long.	Lat.	dp(°)	dm(°)	Lp
in situ	56.2	5.2	27	26.421	44.91	4.2	107.1°E	21.5°N	2.7	4.2	2.6°N
"unfolded"	66.3	-27.3	27	26.404	43.63	4.3	107.7°E	1.6°N	2.5	4.6	14.5°S

McFadden Fold Test:	Observed Value of f^*	Critical Value	
		Level of significance	
		0.05	0.01
in situ	0.24	0.13	0.20
"unfolded"	0.08	"	"

Component I

Limb1: Sites S2,8,10,12
Limb2: Sites S33 to 30,28,36,27,24,23,19 to 17

Limb1 statistics

Mean direction:	D(°)	I(°)	N	R	k	α_{95}(°)
in situ	195.2	7.0	4	3.900	30.03	17.0
"unfolded"	200.8	67.9	4	3.809	15.70	23.9

Limb2 statistics

Mean direction:						
in situ	195.1	15.8	12	11.325	16.30	11.1
"unfolded"	221.2	66.4	12	11.264	14.95	11.6

Overall statistics

Mean direction:							<-----Pole-----|| Long.	Lat.	dp(°)	dm(°)	Lp
in situ	195.1	13.6	16	15.191	18.55	8.8	333.1	28.1°S	4.6	8.0	6.9°S
"unfolded"	216.3	67.0	16	15.045	15.71	9.6	326.5	17.8°N	13.2	15.9	49.7°S

McFadden Fold Test:	Observed Value of f^*	Critical Value	
		Level of significance	
		0.05	0.01
in situ	0.07	0.24	0.39
"unfolded"	0.03	"	"

Notation as previously defined except: R= Vector sum of N unit vectors, dm,dp= semi axes of the oval of 95% confidence for the pole along and perpendicular to the palaeomeridian, Lp= palaeolatitude of site.
Calculated using equation (33), p24: McFadden and Lowes (1981).

6.1 Component H

There appear to be systematic oscillations in direction of component H with stratigraphic position, between sites S34 and S27 and sites S21 and S18. These correlate with higher amplitude oscillations observed by Morris (1973) in laterally equivalent strata. If such variations occur throughout the whole section, it is possible that our sampling has not been adequate to determine mean direction and precision adequately. Moreover the McElhinney test proves inconclusive at the 95 per cent confidence level (Table 2). Despite the fact that Fig. 8 shows the mean directions of component H in each structural unit converging during 'unfolding', the

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test indicates no significant improvement ($p = 0.05$) in the overall grouping of directions from the two units. A marginal increase in the dispersion of site mean directions within each structural unit, emanating from the application of unique 'unfolding' rotations at each site, obscures an improvement in the alignment of the overall directions from each structural unit.

On the other hand McFadden & Jones (1981) have shown that the McElhinney test, applied above, is not strictly valid and have developed a correct test which, it is demonstrated, could show significance in instances which fail the McElhinney test. The McFadden fold test requires the precision parameters for the groups of directions in each structural unit to be the same. This is not the case for either 'unfolded' or 'folded' component H directions from the Salrock Formation (Table 2). The difference in k estimates for each population probably arises from (1) differing proportions of good and poor recorders of the component H remanence (red and green lithologies respectively) sampled in each unit, and (2) more detailed sampling in the upper unit, inevitably yielding more component H directions (20, compared with 7 from the lower structural unit, Table 2). It is therefore reasonable to test the hypothesis that the true mean directions for each unit are not different despite the difference in their precision parameters, as described by McFadden & Lowes, (1981, p. 24).

An approximate test (McFadden & Lowes 1980, equation 33) incorporating an estimate of the ratio of precision parameters was utilised. Marginal significance using this approximate test should be interpreted tentatively. When the test is applied to *in situ* component H results the hypothesis of a common true mean direction for the two populations of site mean directions from each structural unit may be rejected at the 99 per cent confidence level; but when the same test is applied to the 'unfolded' directions the null hypothesis of a common true mean direction cannot be rejected, even at the 95 per cent confidence level. Therefore this test seems to indicate a pre-folding origin for component H, a conclusion that is confirmed in the Clare Island region (see below).

The component of NRM labelled −H, isolated in the metamorphic aureole at site S4 (adjacent to the andesite sill at S3) is antiparallel to the component H directions described above. Hence the CRM (−H) acquired during or shortly after sill intrusion must also pre-date folding.

6.2 Component I

Component I was isolated during relatively few demagnetisation experiments, and only 4 and 12 well determined site level directions of component I were obtained from the stratigraphy lower and upper structural units respectively (Fig. 6a, b, Table 2). The mean directions of component I in each structural unit are shown in Fig. 8 before and after 'unfolding'. McFadden's approximate fold test (McFadden & Lowes 1981, equation 33) was applied to these directions and the null hypothesis of a common true mean direction for the upper and lower parts of the Salrock Formation cannot be rejected either before or after 'unfolding', the level of distortion of the strata being too small to give a significant result with the small number of data (Table 2). The relative ages of remanence I and folding cannot therefore be established using remanence directions from the Salrock samples alone.

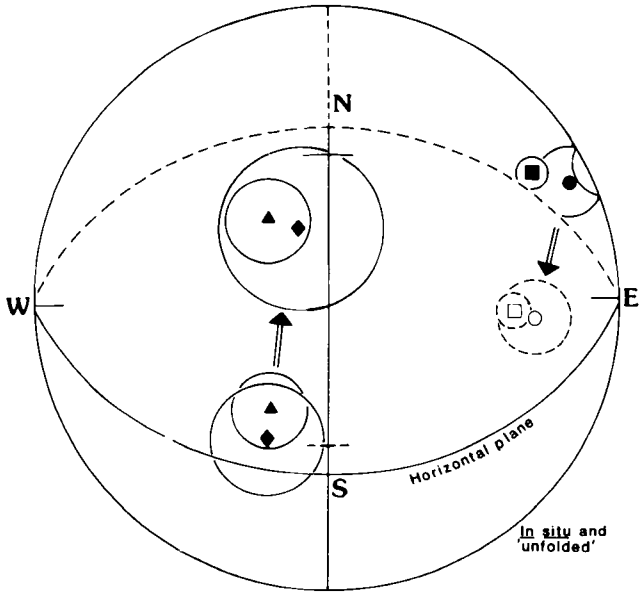


Figure 8. *In situ* and 'unfolded' mean H and I remanence component directions for fold 'limbs' 1 and 2 (which include sites S1 to S14 and S34 to S16 respectively) in the Salrock Formation. Component I: ◆—'Limb' 1, ▲—'limb' 2. Component H: ●—'limb' 1, ■—'limb' 2. Arrows indicate the effect of bedding tilt 'correction' on mean remanence directions. Error circles are of 95 per cent confidence. An equal angle stereographic projection with oblique reference frame has been chosen for clarity. Conventions are as in Fig. 4.

6.3 Irreconcilable results from site S15

High Tb directions from red siltstones at site S15 have uncharacteristic orientations with respect to the attitude of local bedding surfaces and present geographical coordinates.

The most likely cause of local magnetic resetting is thermochemical activity associated with movement on the Salrock thrust, only 88 m from this site. S15 (Fig. 2) is the only site within a cleaved zone beneath the thrust, and it seems probable that the unusual remanence characteristics stem from mechanical, thermal, and chemical effects of cleavage development. The geometry of the remanence directions and thrust related structures excludes mechanical reorientation as a feasible single explanation and a CRM origin is more likely. The fact that component I is preserved at site 15 while component H is distorted, gives palaeomagnetic evidence that the Salrock thrust and associated cleavage developed between the time of origin of component H (which we have argued above is Wenlock) and the remagnetisation that formed component I: the age of I is discussed later.

7 PROGRESSIVE DEMAGNETIZATION: CLARE ISLAND

Although NRM directions are scattered, a clear structure in their distribution is revealed when they are grouped according to lithology (Fig. 9). NRM intensities are

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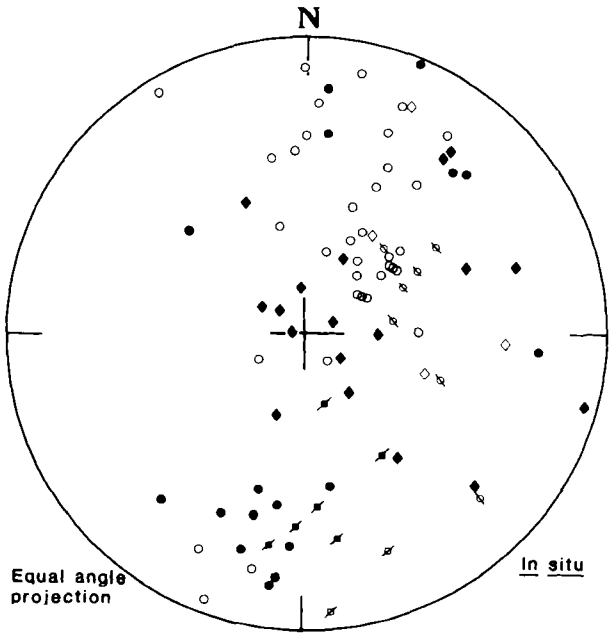


Figure 9. NRM directions of samples from Clare Island in equal angle stereographic projection. Sediments: ●—Bunnamohaun Formation, ⊘—Toormore Group, ■—Allahan Banded Member, ◆—Ignimbrites. Conventions are as in Fig. 4.

commonly between 1 and 20 mA m^{-1}. In all, 80 samples mostly of red sandstones or siltstones were progressively demagnetised. Thermal demagnetisation proved more effective than AF treatment at discriminating vector components of NRM, particularly in redbeds, and was therefore mainly used. A significant increase in bulk susceptibility of samples is observed after heating above 600 °C, yet no systematic spurious magnetisation directions were observed, presumably due to precise field cancellation in the furnace (<5 nT).

Ignimbrites from sites I1 and I2, believed to be in the same flow, responded systematically to both thermal and AF demagnetisation, yet little palaeomagnetically interpretable information was obtained.

The Tb and Hcr ranges of NRM components vary between lithologies, but the order in which the components are removed during AF and thermal treatment remains the same so that the H, I and L notation developed for the equivalent strata in the Salrock Formation can be maintained here.

7.1 Red sediments

Orthogonal plots showing typical demagnetisation characteristics of red sediments are given in Fig. 10(a–e). Most redbed samples yield three components of NRM which have similar Tb stability ranges to H, I and L at Salrock. However vector composition of the NRM in Clare Island redbeds differs from that in the Salrock region (except for red mudstones in the Salrock Formation, Fig. 5c) by (1) a larger component I magnetisation in relation to L and H (the direction of I is consequently better determined) and (2) a larger between-site scatter is component H directions (compare Figs 6a and 11a).

Component L magnetisations are removed by treatments up to 200 °C and, although somewhat dispersed, are directed steeply downwards towards the present Earth's field direction. Scatter in these directions is at least partly a consequence of Tb spectra overlapping with I. However L is confidently regarded as recent VRM: petrological analysis reveals coarse MD magnetite which with other opaque minerals such as goethite may contribute to component L.

Component I has Tb between 150 °C and 570–650 °C. This is the broadest stability range of the three components, yet component I rarely constitutes as large a part of the total NRM as component H. The site mean directions of component I, which are all of opposite polarity to H, are given in Fig. 11(a) and Table 3. The magnetic properties of component I are consistent with host mineral phases of magnetite and haematite. Magnetite is observed as detrital grains and may also occur as inclusions within other detrital mineral phases in the redbeds.

Component H has a narrow, unique Tb spectrum between 550 °C and 680 °C. It is usually the largest component of NRM, and commonly has a well defined normal polarity with respect to the generally accepted apparent polar wander path for the British Isles. The magnetic properties of component H indicate that it is carried predominantly by haematite grains which may be identified under the microscope in the form of early diagenetic specular or colloidal intergranular masses and grain coatings and, to a lesser extent, in the form of original detrital grains.

7.2 Ignimbrites

Sites I1 and I2, from an ignimbrite flow, do not yield a remanence comparable with the southerly and shallow direction (component I) observed elsewhere (Fig. 10f). Component L is present but is scattered in direction, persisting up to 450 °C, well into the usual Tb range of component I in the redbeds. A high Tb component (520 °C to the Curie point of haematite) is observed in a few samples, and is similar in direction (relative to bedding) to component H in the redbeds. These directions are dispersed, yet they do confirm the presence of the remanence H in approximately contemporaneous but contrasting lithologies.

8 FOLD TESTS IN THE CLARE ISLAND REGION

The folds sampled in the Bunnamohaun Formation (above the Owenmore thrust) have approximately horizontal axes with open hinges such that there is a maximum variation in bedding orientation of *ca.* 70° between sampling sites (Table 3). Site B8 (below the Owenmore thrust) did not yield a component H direction and is therefore only included in a fold test for component I. Limited exposure restricted sampling to one site per fold limb and some uncertainty lies not only in the number of fold limbs within unexposed tracts between sites, but also in the way-up of some steeply inclined strata. The conventional 'unfolding' method using rotations about bedding strike directions (Graham 1949) is applied to all site level directions from the Bunnamohaun Formation and Allahan Banded member.

Site T1 is located in the relatively tightly folded Toormore Group south of the Kill fault, on the limb of a fold with axis

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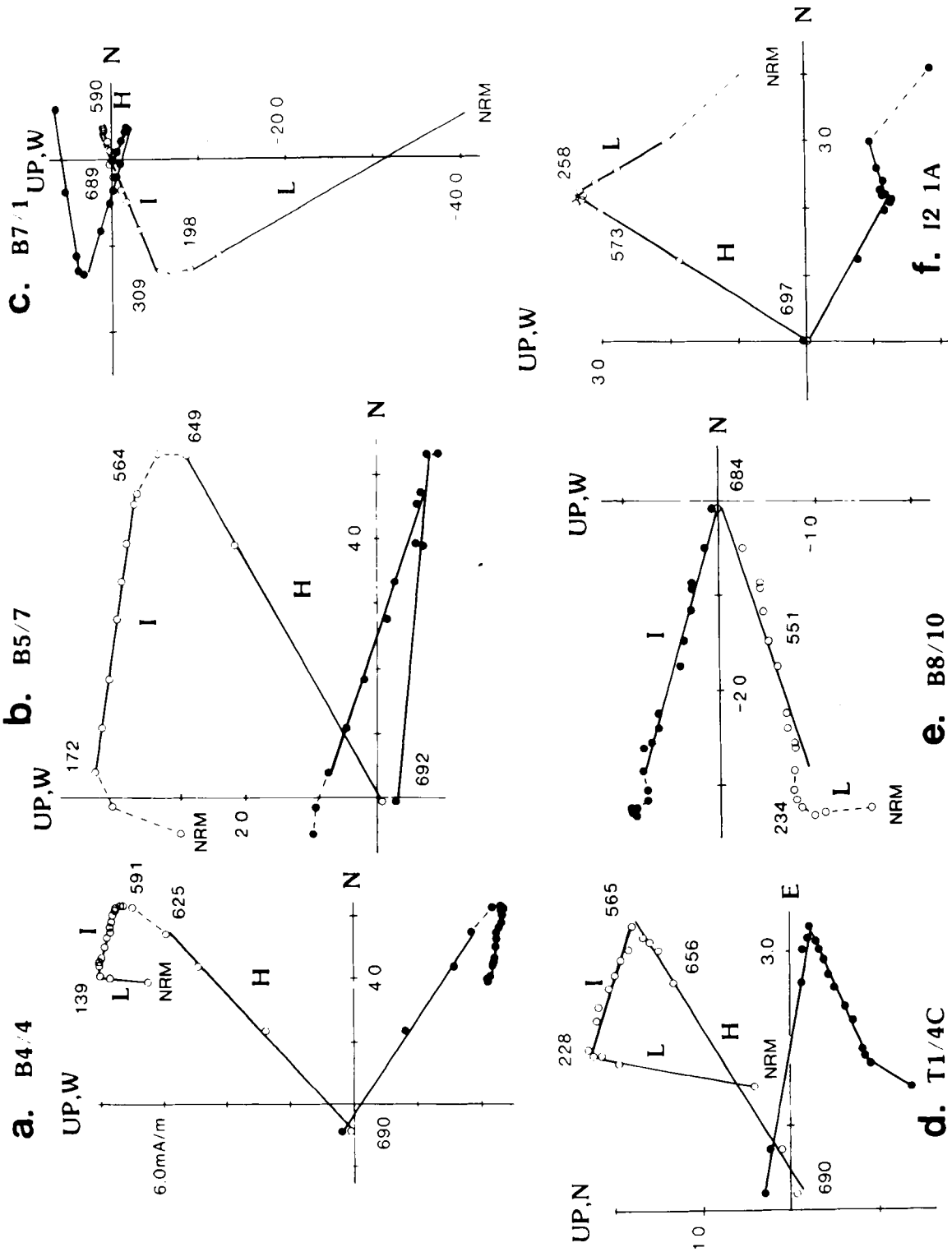


Figure 10. Orthogonal plots of the results of progressive thermal demagnetisation experiments on samples from Clare Island. (a) to (e) red sediments, (f) ignimbrite. Symbols and labels are as in Fig. 5.

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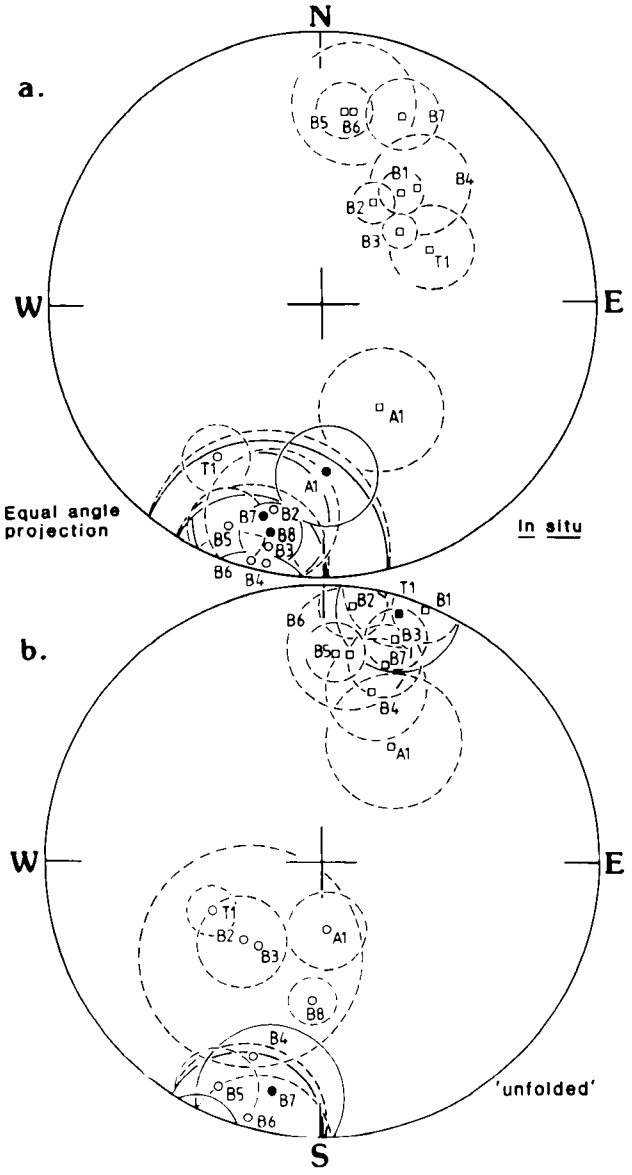


Figure 11. Site level mean H(■) and I(●) NRM component directions for the Clare Island succession in equal angle projection. (a) *In situ* directions. (b) 'Unfolded' directions. Error circles are of 95 per cent confidence. Site numbers are indicated. Conventions are as in Fig. 4.

plunging at 46° towards 090°. Here the restoration of strata to their presumed original attitude involves 'unplunging' followed by residual 'unfolding'. 'Folded' (*in situ*) and 'unfolded' site level directions from the Clare Island succession are shown in Fig. 11(a) and 11(b), respectively.

Although, as already mentioned, the straightforward McElhinny (1964) fold test is not statistically rigorous, where it does give indication of statistical significance, the more complicated McFadden test will do so also. For this reason the McElhinney test is applied to establish the ages of remanences H and I from the Clare Island succession relative to folding. The results of this test indicate a prefolding origin for component H (consistent with Salrock Formation results) and a post-folding origin for component I at the 95 per cent confidence level (Table 4).

Rotations on the Owenmore thrust (Philips 1974) or on the Kill fault are not required to reconcile component H directions from either side of the respective major dislocations, and such rotations are therefore regarded as small or absent. Component H directions from sites where no way-up criteria are evident are all reconcilable by 'unfolding' of local strata on the assumption that they are the right way up.

9 REGIONAL RECONCILIATION OF OVERALL MEAN DIRECTIONS

9.1 Component H

Overall mean remanent magnetization directions for the two study areas, both *in situ* and 'unfolded', are depicted in Fig. 12(a). The overall 'unfolded' component H directions for the Clare Island and Salrock successions are different despite their otherwise similar magnetic properties and the similar age and character of the host lithologies. The 'unfolded' orientation of the component H remanence from Clare Island is removed by approximately 45° anticlockwise in declination from that in the Salrock Formation. The two Formation mean directions make similar azimuthal angles with the local structural trends, and when those structural trends are rotated into alignment the circles of 95 per cent confidence on the palaeomagnetic directions then overlap (Fig. 12b). Hence the widely separated *in situ* component H directions from the two regions are brought into good within-region agreement upon complete 'unfolding' about local fold axes, and into inter-regional alignment when local strike directions are brought into parallelism. This palaeomagnetic compartmentation indicates rotation of internally uniform blocks which are larger than the sampling spread in either region (*c.* 4 km) and smaller than the distance between the two regions (*c.* 25 km).

It is clearly crucial to establish which of the two overall component H remanence directions, if either, is a faithful record of the palaeofield direction relative to the rest of Britain at the time of remanence acquisition, and which has subsequently been rotated. Most of the major faults in the Salrock area accomodated much larger displacements in the Ordovician than in the Silurian period (Ryan, personal communication). Stratigraphic relations across the Salrock Fault (at the top of the Salrock Formation) show little if any rotational effect beyond its immediate vicinity (i.e. component H at site S15) and it is reasonable to conclude that the folded Salrock strata have remained unaffected by local block rotations through post-Silurian times. There is, however, clear potential for considerable sinistral strike-slip displacements on faults in the Clare Island (Clew Bay) area well into Devonian times (Graham 1981; also see Soper & Hutton 1984; Dewey & Shackleton 1984), which provide the most likely conditions for localised anticlockwise block rotations. Therefore, it seems probable that it is the declination from Clare Island which is anomalous and that the declination at Salrock is more representative of the

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Table 3. Clare Island: site mean directions of components H and I (*in situ*, in approximate stratigraphic order)

Site	Lithology	Strike, Dip(°)	Component H: D(°)	I(°)	N	k	α_{95}(°)	Component I: D(°)	I(°)	N	k	α_{95}(°)	
Bunnamohaun Siltstone Fm.													
B8		140 57SW	-	-	-	-	-	192.9	8.9	6	77.0	7.7	
B6	Predominantly Red	030 7SE	9.5	-19.1	4	30.2	17.0	195.5	-1.2	5	17.1	19.1	
B5	Siltstones	023 10SE	6.9	-19.7	6	74.6	7.8	203.1	-7.2	6	48.5	9.7	
B4		037 41SE	39.8	-32.7	5	24.3	15.8	192.4	-1.6	4	4.1	NS	
B7		020 19SE	23.4	-17.0	6	46.2	9.9	195.7	12.3	6	14.4	18.3	
B3		072 62S & 067 58S	47.6	-48.0	6	114.5	6.3	192.6	-5.5	3	15.1	32.9	(in situ)
			18.0	-10.4	6	76.4	7.7	216.2	-49.1	3	11.3	38.5	("unfolded")
B2		061 56S	27.3	-45.6	5	94.3	7.9	193.4	-14.5	5	21.2	17.0	
B1		072 50S	35.8	-37.4	6	79.8	7.6	-	-	-	-	-	
I1	Ignimbrite	Approx. same as B1											
I2	Ignimbrite	063 76S											
Allahan Banded Member													
A1	Red Sandstone	092 90S	151.9	-42.9	3	31.4	22.4	179.0	26.7	6	20.0	15.4	
Toormore Grp.													
T1	Red Siltstone	088 72S & 092 68S	63.8	-42.6	6	21.7	14.7	214.6	-21.7	6	47.9	9.8	(in situ)
		Fold axis: 090 46	46.9	4.4				246.4	-56.1				("unfolded")

Notation as previously defined

Table 4. Clare Island: group of sites level statistics and fold tests

Component H

Incorporating directions from sites T1,A1,B1 to B7

Overall statistics Mean direction:	D(°)	I(°)	N	R	k	α_{95}(°)	Pole: Long.	Lat.	dp(°)	dm(°)	Lp
in situ	35.9	-39.8	9	7.693	6.12	22.7	136.9°E	7.6°N	16.3	27.2	22.6°S
"unfolded"	14.0	-14.2	9	8.708	27.42	10.0	154.3°E	27.9°N	5.2	10.3	7.2°S

McElhinny Fold Test:	Observed F-ratio (k2/k1)	Critical F-ratio: Level of significance 0.05	0.01
k2: 27.42	4.48	2.33	3.37
k1: 6.12			

Component I

Incorporating directions from sites T1,A1,B2 to B8

Overall Statistics Mean direction:	D(°)	I(°)	N	R	k	α_{95}(°)	Pole: Long.	Lat.	dp(°)	dm(°)	Lp
in situ	195.5	-0.5	9	8.641	22.30	11.1	331.0°E	34.9°S	5.6	11.1	0.3°N
"unfolded"	201.8	-23.5	9	7.933	7.50	20.1	319.1°E	45.0°S	11.4	21.4	12.2°S

McElhinny Fold Test:	Observed F-ratio (k1/k2)	Critical F-ratio: Level of significance 0.05	0.01
k2: 7.50	2.97	2.33	3.37
k1: 22.30			

Notation as previously defined

regional Silurian meridian. Also the Salrock result agrees in declination with British palaeofield directions of Silurian and Lower Devonian age (Table 5: Comrie Complex, Tortworth Inlier, Anglo–Welsh Cuvette).

There has recently been some uncertainty expressed regarding the nature and extent of Silurian deformation in the rocks to the south of Killary Bay Little (Fig. 2; Williams, personal communication), however there are no anomalies in the direction of observed ancient remanences and there is no palaeomagnetic reason to suspect block rotation in that area.

9.2 Component I

The Overall *in situ* component I direction for the Clare Island succession has been shown to post-date folding, and is similar in direction to I for the Salrock Formation (for which, it will be recalled, there is no conclusive fold test (Fig. 8, Table 2). Consequently both directions are regarded as estimates of the same post-folding palaeofield direction, which must also post-date local block rotation.

Folding events which disperse component H directions in both regions are generally regarded as late Silurian–early Devonian in age (Dewey 1963). Block rotation could have

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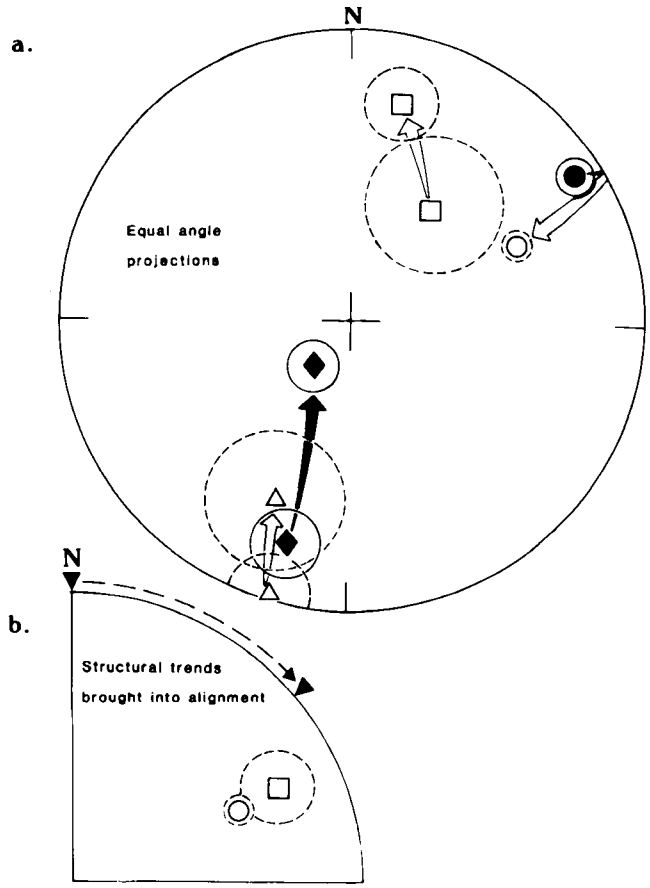


Figure 12. *In situ* and structurally 'corrected' overall mean NRM component directions for the Salrock and Clare Island successions. Salrock: ●—H component, ◆—I. Clare Island: ■—H, ▲—I. Arrows indicate the effect of structural 'corrections' on overall remanence directions. (a) *In situ* directions and directions 'corrected' for local folding. (b) The 'unfolded' H component direction from the Salrock Formation and the 'unfolded' Clare Island result after anticlockwise rotation of 45°, a rotation which brings regional structural trends into parallelism.

continued well into Devonian times driven by strike-slip faulting in the Clew Bay area. It seems probable therefore that component I, which post-dates block rotation, is post-early–middle Devonian in age.

10 REMAGNETISATIONS

A younger age limit of component I may be estimated by comparison between the results presented in this paper and previous results from Britain. I directions are similar to the listed directions of Upper Devonian–Lower Carboniferous age (Table 5). It is reasonable to suppose that component I was imposed in Devonian to Carboniferous times, after tectonism had ceased.

Evidence for the age of component I can also be derived from diagenetic studies. Reddened strata commonly exhibit emergent sedimentary structures such as desiccation polygons. A few green sandstones, in which the sole opaque mineral phase identified is magnetite, contain reddened rip-up clasts which themselves contain at least two readily identified and palaeomagnetically significant opaque mineral phases: magnetite and haematite. This provides clear evidence for the early formation of specular or colloidal haematite (the reddening agent) during diagenesis, probably during periods of temporary emergence and oxidation of original detrital magnetite. Some detrital haematite is also observed.

Since the younger component I is predominately carried by magnetite it seems that I was produced by either thermal resetting of pre-existing detrital magnetite grains, or by micro-exsolution in the detrital magnetite (though this has not been conclusively identified) or a combination of the two processes, occurring some time after early diagenesis.

If the origin of the I remagnetisation were predominately a thermal one, then the timing and depth of subsequent burial (the most likely cause of large scale region-wide temperature elevation) would provide supporting evidence for age and mode of remagnetization. Burial during the later stages of the Caledonian orogeny followed by post-Caledonian uplift, cooling and remanence blocking could have provided suitable conditions for the formation of viscous partial TRM (VPTRM). However, the maximum Tb of I (*c.* 600 °C) is far in excess of that which could have been acquired by VPTRM, even taking the time-factor into account. Hence I is more likely to be predominantly the result of chemical resetting.

Sedimentary rocks from sites A1 and B8 provide an even firmer example of a CRM origin of I. The two sites lie within tracts of altered sediment adjacent to the Leck fault and Owenmore thrust respectively. At site B8 the secondary magnetisation (I) is stable up to 675 °C, spanning the normal Tb ranges of both I and H (Fig. 10e). At site A1 I is stable up to 620 °C and component H is poorly defined. At these two sites the intermediate Tb component (including high Tb range) is at least in part a CRM, probably acquired during fluid migration and consequent alteration in the vicinity of the then active dislocations. Elevated temperatures adjacent to the dislocations must not be discounted as a contributor to the remagnetisation processes operating. The CRM direction is indistinguishable from that of the remagnetisation elsewhere, and its blocking is likely to have been approximately synchronous throughout the area, dating an episode of thermochemical activity associated with the two faults.

11 COMPARISON OF THE PRIMARY PALAEOFIELD DIRECTION IN W. IRELAND WITH BRITISH RESULTS

The comparison here is made between the W. Irish Silurian palaeofield direction (the 'unfolded' component H direction from the Salrock Formation) and the estimates of Silurian palaeofield directions from previous work in Britain listed in Table 5. It has been established on local structural grounds that the overall 'unfolded' component H direction from the Salrock Formation is likely to represent a faithful record of the Silurian palaeofield direction in W Ireland. Similarity in field azimuth between British and Irish results (this study: Salrock Formation 'unfolded' H direction, Table 2) discounts rotations of this area as an even larger block, thus extending the applicability of the Salrock result beyond W Ireland.

There is fair agreement between the inclinations of the

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Table 5. Selected Palaeozoic pole positions from Britain and Ireland

Period/ Tectonic/Rock Unit	<---Remanence -||<------- Pole(°)-------|| D(°) I(°) α_{95}(°) lat.long. dp/dm, A_{95}	Lp.RC	age	pole no./source reference
SILURIAN				
South of Iapetus				
Somerset & Gloucester Lavas	243 +48 14 8.1 308.7 11.5/17.7	29S D	(U. Llandovery)	15/131 Piper (1975) X
Tortworth Inlier, Gloucestershire	216 +34 - 9.0 287.0 9/16	19S D	(L. Sil.)	4/358 Morris et.al.(1973) X
Dolerite Dyke Swarm Anglesey	53 -30 - -7.0 310.0 9.7	16S	(U. Sil./L. Dev.)	Piper (1976) X
South of GGF, North of Iapetus				
Arrochar Complex and Aureole	213 +37 5 -8.0 324.0 4/6	21S C	(418ma.)	12/139 Briden (1970) X
Garabal Hill-Glen Fyne Complex	32 -42 24 -5.0 326.0 18/30	24S C	(U.Sil./L. Dev.)	12/138 Briden (1970) X
Lorne Plateau Volcanics	40 -49 6 2.0 321.0 5/8	30S B	(U.Sil./L.Dev.)	14/377 Latham & Briden (1975) X
Comrie Complex	75 -30 6 -6.0 287.0 4/7	16S	(408Ma.)	Turnell (1982) X
North of GGF				
Caledonian Dolerites	8 -32 10 -14.7 347.5 6.4/11.4	17S	(435ma.)	Esang & Piper (1984) X
Caledonian Microdiorites	9 -29 6 -16.5 346.4 3.8/6.9	11S	(435ma.)	Esang & Piper (1984) X
Borrolan Ledmorite	344 -37 7 -10.0 19.7 5/7.3	21S	(430ma.)	Turnell & Briden (1983) X
Ratagan Complex	8 -32 8 -15.0 347.0 5/10	17S	(415ma.)	Turnell (1982) X
Moine Meta-sediments (High Tb)	-6.0 313.0 6	11S	(U. Sil.)	Watts (1982) X
Moine Meta-sediments (Int. Tb)	-1.0 309.0 9.5	16S	(U. Sil.)	Watts (1982) X
DEVONIAN				
South of Iapetus				
Anglo-Welsh Cuvette	66 -38 13 3.0 298.0 12/14	21S B	(L.Dev.)	8/126 Chamalaun & Creer (1964)
Hendre & Blodwell	189 +9 15 -32.2 346.8 14.9	5S D	(384ma.)	16/162 Piper (1978)
South of GGF, North of Iapetus				
O.R.S Lavas, Midland Valley	41 -40 6 -4.0 320.0 4/7	23S B	(L.Dev./U.Sil.)	14/379 Sallomy & Piper(1973)
Cheviot Granite	207 +48 17 4.0 337.0 17	29S D	(L. Dev.)	14/371 Thorning (1974)
Cheviot Lavas	49 -53 15 11.0 320.0 18	34S D	(L. Dev.)	14/370 Thorning (1974)
O.R.S Lavas Scotland	46 -54 8 10.0 321.0 8/12	35S	(L. Dev.)	8/125 Stubbs (1958)
Jedburgh O.R.S.	188 +22 9 -23.0 349.0 5/9	11S	(U. Dev.)	Nairn (1960)
Derry Bay Felsite Ireland	192 +28 35 -21.0 338.0 21/38.3	15S	(U. Dev.)	15/127 Morris (1976)
North of GGF				
Foyers Plutonic Complex	004 -6 18 -29.0 340.0 9/18	3S D	(L. Dev.)	14/380 Kneen (1973)
Eday Lavas & Baked Contact	191 +41 8 -8.0 347.0 -	23S	(M. Dev.)	Robinson (1985)
John O'Groats Sandstone	188 -41 4 -54.0 344.0 -	23N C	(M.Dev.)	Storetvedt & Carmichael (1979)
CARBONIFEROUS AND PERMIAN				
South of Iapetus				
Bristol Dev. and Carb. Sediments	196 +9 10 -32.0 338.0 5/10	5S	(U.Dev/L.Carb.)	14/359 Morris et al(1973)
Carb. Limestone Ireland	188 -13 10 -43.0 341.0 7/13	7N	(Tournasian)	15/121 McAulay & Morris(1971)
Combined Carb. Limestones	201 -15 12 -40.5 328.4 8.9	8N	(Carb.)	14/358 Turner & Tarling (1975)
Whin Sill	194 -22 5 -44.0 339.0 3/5	11N	(U.Carb.)	11/77 Storetvedt & Gidskehaug (1969)
Wackerfield Dyke	186 -26 3 -49.0 349.0 2/3	14N	(U. Carb.)	14/350 Tarling et al (1973)
Exeter Lavas	198 -25 7 -50.0 329.0 4/8	13N	(L. Perm.)	9/89 Zijderveld (1967)
	188 -13 7 -46.0 45.0 4/7	7N	(L. Perm.)	9/90 Cornwell (1967)
South of GGF, North of Iapetus				
Kinghorn Lavas	16 -28 8 -18.0 341.0 5/9	15S	(L. Carb.)	8/118 Wilson & Everitt (1963)

RC = Briden's reliability classification of the palaeomagnetic data (Briden & Duff, 1981).
Pole numbers 8/, 9/, 11/, 12/ from McElhinny (1968a,b; 1970; 1972a).
Pole numbers 14/, 15/, 16/ from McElhinny & Cowley (1977; 1978; 1980).

Iapetus = Trace of Iapetus Suture
GGF = Great Glen Fault
Sil. = Silurian
Dev. = Devonian
Carb. = Carboniferous
Perm. = Permian
L.= Lower, M.= Middle, U.= Upper

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Salrock Formation and Clare Island results, which correspond with an average palaeolatitude of 11.2°S. Palaeolatitudinal estimates based on Silurian results from Britain north of the Great Glen fault are similar to this (Table 5). Surprisingly, palaeolatitudes for the rest of Britain based on results from both north and south of the Iapetus Suture are all further south than the Irish result. The difference is of the order of 5°–10° (Table 5). Ninety-five per cent confidence limits on the palaeolatitudinal estimates used here (marked with X in Table 5) vary between ±6° and ±30°. When this uncertainty is combined with uncertainty in the two Irish results (±8°–±20°, Tables 2 and 4) it becomes clear that individual differences of 5°–10° in palaeolatitude have no statistical significance. The width of the Iapetus Ocean in Silurian times may, however, be more confidently estimated by comparing earlier and later sections of the apparent polar wander path for the major tectonic units in Britain, while calling on other lines of evidence such as minimum ocean closure rates, faunal provinciality and cleavage development during plate collision. Width estimates of *c.* 1000 km for Ordovician times (Briden, Turnell & Watts 1984), 900 km for Silurian times (Cocks, McKerrow & Leggett 1980), and complete closure some time in the Devonian period (Soper & Hutton 1984; Livermore *et al.* 1985; Ziegler 1982) are consistent with a closure rate, supported by structural evidence from the Southern Uplands of Scotland (McKerrow Leggett & Eales 1977), of 2 cm yr^{-1}.

The Wenlock result from the Salrock region does not require clockwise rotation of the Connemara region to bring it into alignment with equivalent British directions, as was thought necessary by Morris (1973) in reconciling his result with those from Britain, On the contrary the Salrock result provides a minimum age for the postulated clockwise rotation of Connemara (Leake *et al.* 1983) of 415–422 Ma (Holland 1985): the likely age of deposition and diagenesis of the Salrock Formation.

Hence comparison of the overall mean directions H ('unfolded') and I *in situ* from the Salrock and Clare Island regions with British results helps define a sequence of four separate events which succeed the proposed clockwise rotation of Connemara. These are (1) sediment diagenesis and haematite remanence acquisition (H) on the northern margin of the Iapetus Ocean, in Silurian times, (2) Late Caledonian folding (associated with continued closure of Iapetus) probably late in Silurian or early in Devonian times, (3) block rotation of Clare Island, conceivably driven by oblique plate convergence and (4) partial remagnetisation (I) of probable late Devonian or Carboniferous age.

12 CONCLUSIONS

It has been shown that Silurian sedimentary rocks sampled in the vicinity of Salrock village in NW Galway and on Clare Island in Co. Mayo carry a primary or near-primary diagenetic magnetisation which corresponds to a palaeomagnetic pole position at longitude 287.7°E, latitude 1.6°S. The emplacement of andesite and lamprophyre sills caused local chemical remagnetisation in the Salrock Formation, before a late Caledonian tectonic phase folded the strata.

The Wenlock sedimentary succession on Clare Island is probably correlative with the Salrock Formation. A local fold test there reveals a primary or near-primary component of NRM removed in declination by 40°–50° anticlockwise from the Salrock result. Also, structural trends in the two regions differ in azimuth by approximately this amount, which suggests anticlockwise block rotation of Clare Island relative to the Salrock area, through 40°–50°. Strike-slip displacements on the Irish extension of the Highland Boundary fault (Max & Riddihough 1975), in Clew Bay, producing a sinistral shear couple might have enabled such a rotation. However, it is somewhat surprising that a rotation of this size could occur without causing extensive deformation within the rotated block. Nevertheless similar rotations have been identified by Wells & Coe (1985) in the Western margin of the North American plate where the interiors of some large blocks have remained internally coherent. A partial chemical remagnetisation of the Clare Island and Salrock successions occurred after relative movement between the two regions had ceased, after mid-Devonian times. Further detailed palaeomagnetic analysis of Palaeozoic rocks in the Clew Bay area and to the northwest in the Ox Mountains inlier is required to define better the extent and timing of block rotation in the area.

The Silurian palaeomeridian deduced from the Salrock Formation cannot be distinguished from that of Britain at the same time. Consequently the proposed clockwise rotation of Connemara (Leake *et al.* 1983) must pre-date the acquisition of this remanence. Ordovician palaeomagnetic results from the Connemara Gabbro appear to show a clockwise rotation in excess of 30° (Robertson 1988), providing an older age limit for rotation. Other lines of evidence for the age of clockwise rotation in Connemara, with which the palaeomagnetic evidence presented here is consistent, are discussed by Leake (1986).

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